Intro to Data Science HW 8

Copyright 2025, Jeffrey Stanton, Jeffrey Saltz, Christopher Dunham, and Jasmina Tacheva

```
# Enter your name here: Sofia Quintero
```

USE OF ARTIFICIAL INTELLIGENCE: While you are allowed to use generative AI as a learning tool for this assignment (e.g., to help with brainstorming, to see how to code a certain concept, to help identify flaws in reasoning, or to spot confusing or underdeveloped code segments), it must not be used to input homework questions, problems, or instructions directly, and you may not copy and paste any AI-generated content into your work. By submitting this assignment, you confirm that all the work is your own.

Attribution statement: (choose only one and delete the rest)

```
# 1. I did this homework by myself, with help from the book and the professor.
```

This homework assignment introduces linear predictive modeling through the use of multiple regression, which is a tool for assessing the relationship between a set of predictor variables and an outcome variable. The term "multiple regression" originated from an early scientific observation of a phenomenon called "regression to the mean," but today it is simply a name for using linear modeling to analyze data.

In this homework, we will be predicting house prices using several predictor variables. To do so, we will use the publicly available "House Sales in King County, USA" dataset, which can be found at the following URL:

https://data-science-intro.s3.us-east-2.amazonaws.com/house-prices.csv

To begin, load the dataset into a pandas dataframe named "house_df". Next, use the relevant function(s) to generate a summary of the data. Add a comment explaining the number of variables and observations in the dataframe and show the function(s) you used to generate this information.

```r
install.packages("tidyverse", repos = "http://cran.us.r-project.org")
```

```
##
## The downloaded binary packages are in
##  /var/folders/rm/s4vv52kd55g34xzsxxqn9p2c0000gn/T//RtmpGHibBX/downloaded_packages
```

```r
library("tidyverse")
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
house_df <- read_csv("https://data-science-intro.s3.us-east-2.amazonaws.com/house-prices.csv")
```

```
## Rows: 128 Columns: 8
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (2): Brick, Neighborhood
## dbl (6): Home, Price, SqFt, Bedrooms, Bathrooms, Offers
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
glimpse(house_df)
```

```
## Rows: 128
## Columns: 8
## $ Home         <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17~
## $ Price        <dbl> 114300, 114200, 114800, 94700, 119800, 114600, 151600, 15~
## $ SqFt         <dbl> 1790, 2030, 1740, 1980, 2130, 1780, 1830, 2160, 2110, 173~
## $ Bedrooms     <dbl> 2, 4, 3, 3, 3, 3, 3, 4, 4, 3, 3, 2, 3, 3, 4, 4, 3, 3, 2, ~
## $ Bathrooms    <dbl> 2, 2, 2, 2, 3, 2, 3, 2, 2, 3, 2, 2, 2, 3, 3, 2, 3, 3, 2, ~
## $ Offers       <dbl> 2, 3, 1, 3, 3, 2, 3, 2, 3, 3, 3, 2, 4, 5, 4, 1, 4, 4, 1, ~
## $ Brick        <chr> "No", "No", "No", "No", "No", "No", "Yes", "No", "No", "N~
## $ Neighborhood <chr> "East", "East", "East", "East", "East", "North", "West", ~
```

```r
# 128 rows and 8 observations
```

B. In the subsequent analysis, **Price** will serve as the outcome variable, while the size (in square feet), number of bedrooms and bathrooms, and whether the house is made of brick or not will serve as predictors. Include a comment briefly describing the outcome variable.

Include a second comment to explain the difference between the outcome variable and the predictors in linear regression.

```r
# Comment1: The outcome variable 'Price' is a double describing the price each
# house in the data set. It will observed how this variable changes based off
# the specified predictors.
```

```r
# Comment2: In linear regression, the predictor variable influences the outcome
# variable. The relationship the two variables have is seen in the linear model
# created. The predictor can be described
```

C. Before proceeding with the analysis, it is essential to inspect the outcome and predictor variables for any missing values. Show the code you used to check for missing values in these variables.

Explain the output of the code in a comment - what does it say about the number of missing values in each variable?

```r
nrow(subset(house_df, is.na(house_df$SqFt)))
```

```
## [1] 0
```

```r
nrow(subset(house_df, is.na(house_df$Bedrooms)))
```

```
## [1] 0
```

```r
nrow(subset(house_df, is.na(house_df$Bathrooms)))
```

```
## [1] 0
```

```r
nrow(subset(house_df, is.na(house_df$Brick)))
```

```
## [1] 0
```

```r
# There are no missing values in any of the variables
```

D. Next, it's essential to ensure that all four predictor variables listed in Step B are coded as numerical values. If any predictor is not already in numerical form, it must be converted. The only exception is the "Brick" variable, which has only two values: "Yes" and "No." The code below demonstrates how to convert it into a binary format — be sure to include a comment explaining what the code does, and provide the code you used to check the variable types of the other predictors.

```r
house_df$Brick <- as.numeric(as.factor(house_df$Brick), levels = c(0, 1))
# Comment: Turns 'no' into '1' and 'yes' into '2' in order to work with
# numerical values
```

```r
# Check the format of the rest of the predictors:
typeof(house_df$SqFt)
```

```
## [1] "double"
```

```r
typeof(house_df$Bathrooms)
```

```
## [1] "double"
```

```r
typeof(house_df$Bedrooms)
```

```
## [1] "double"
```

Once the conversion is complete, make sure to check again for any missing values. If any missing values are present, an appropriate strategy such as interpolation() from the **imputeTS** package we have used in previous HW assignments should be used.

```r
nrow(subset(house_df, is.na(house_df$SqFt)))
```

```
## [1] 0
```

```r
nrow(subset(house_df, is.na(house_df$Bedrooms)))
```

```
## [1] 0
```

```r
nrow(subset(house_df, is.na(house_df$Bathrooms)))
```

```
## [1] 0
```

```r
nrow(subset(house_df, is.na(house_df$Brick)))
```

```
## [1] 0
```

E. To explore the relationship between each predictor and the outcome variable, we will create four visualizations using ggplot. In each plot, plot the outcome variable, price, on the Y-axis, and the respective predictor variable on the X-axis. It's up to you to decide on the most informative way of visualizing the relationship between the outcome variable and each predictor - e.g. by using scatterplots or boxplots. Don't forget to clearly label the X- and Y-axis using xlab() and ylab() and add a title to each plot.

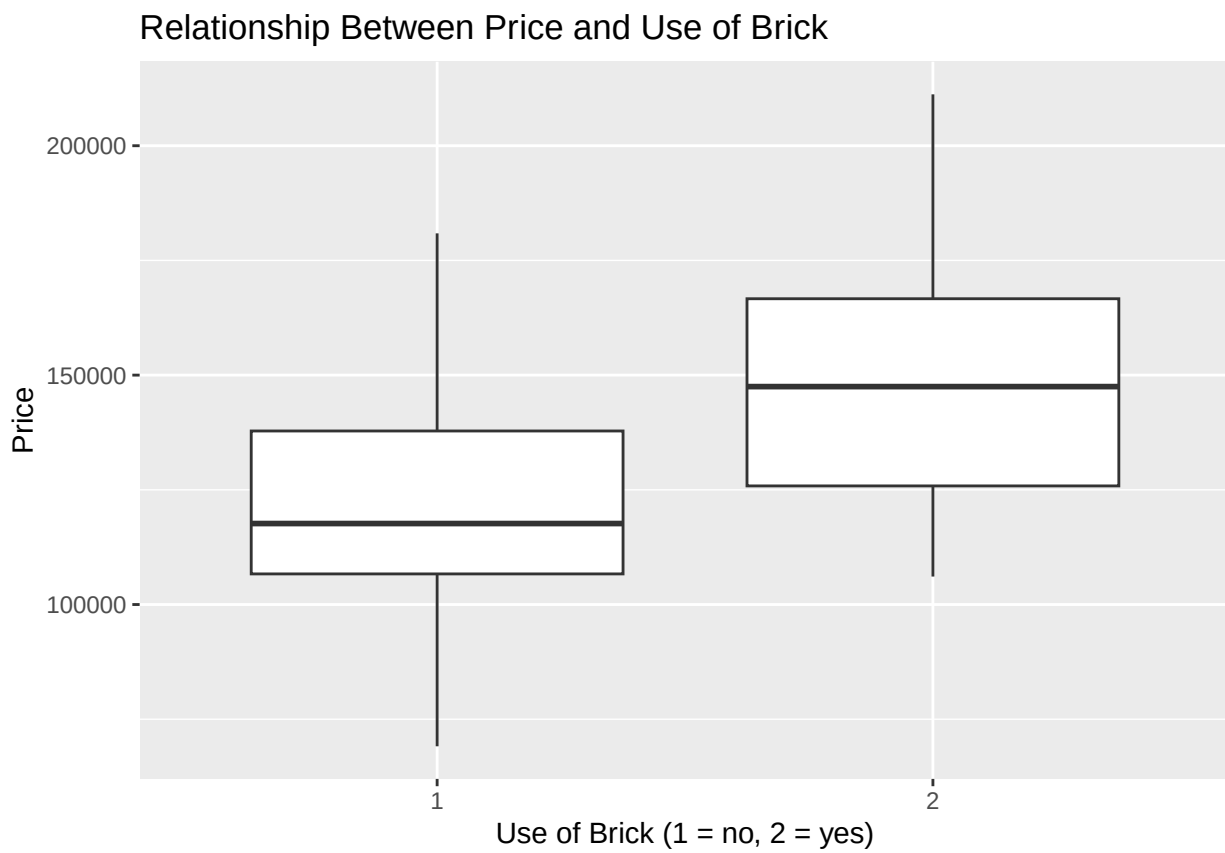
In a comment, explain your decision to use a particular kind of graph and describe each plot, indicating whether there appears to be a linear relationship between the outcome variable and the respective predictor variable.

```r
install.packages("ggplot2", repos = "http://cran.us.r-project.org")
```

```
##
## The downloaded binary packages are in
##  /var/folders/rm/s4vv52kd55g34xzsxxqn9p2c0000gn/T//RtmpGHibBX/downloaded_packages
```

```r
library("ggplot2")
```

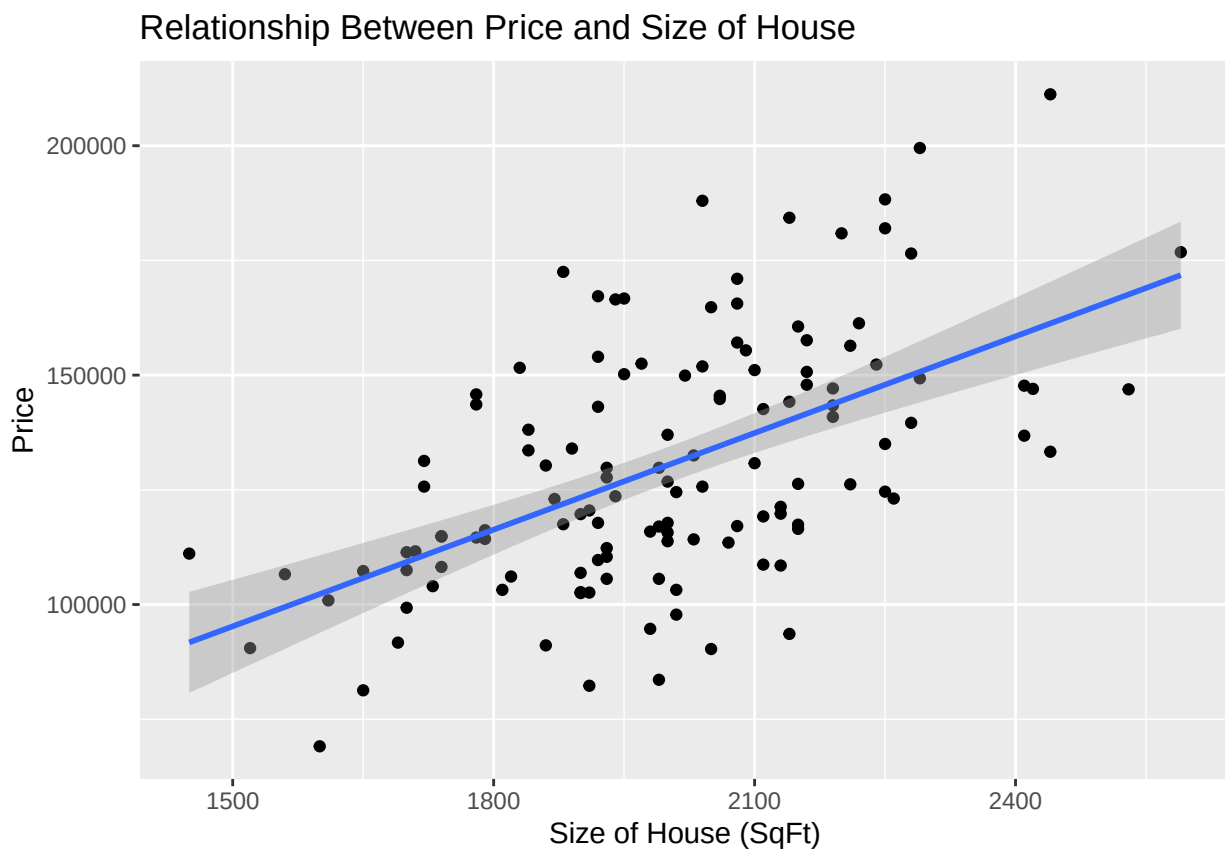
```r
# relationship between price and use of brick

brickPlot <- ggplot(house_df, aes(x = (as.factor(house_df$Brick)), y = Price)) +
    geom_boxplot() + xlab("Use of Brick (1 = no, 2 = yes)") + ggtitle("Relationship Between Price and Us
brickPlot
```




```
# I opted for a boxplot to visualize this relationship due to there only two
# x-values. This allows for comparison of prices between the predictors based
# off statistical analysis, While linear relationship can't be determined, the
# plot does show that houses which use brick are priced higher.
```

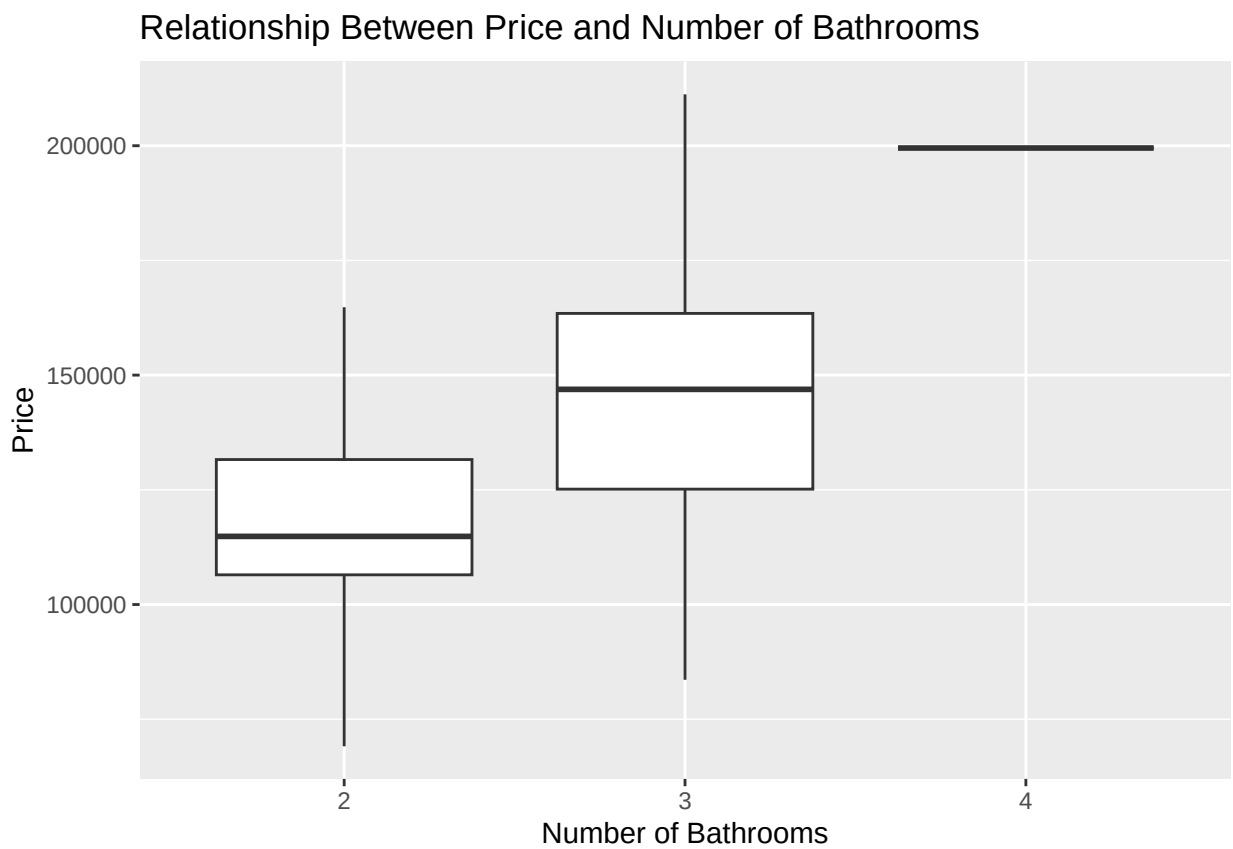
```
# Relationship Between Price and Size of House

sizePlot <- ggplot(house_df, aes(x = SqFt, y = Price)) + geom_point() + xlab("Size of House (SqFt)") +
    ggtitle("Relationship Between Price and Size of House") + geom_smooth(method = lm)
sizePlot
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



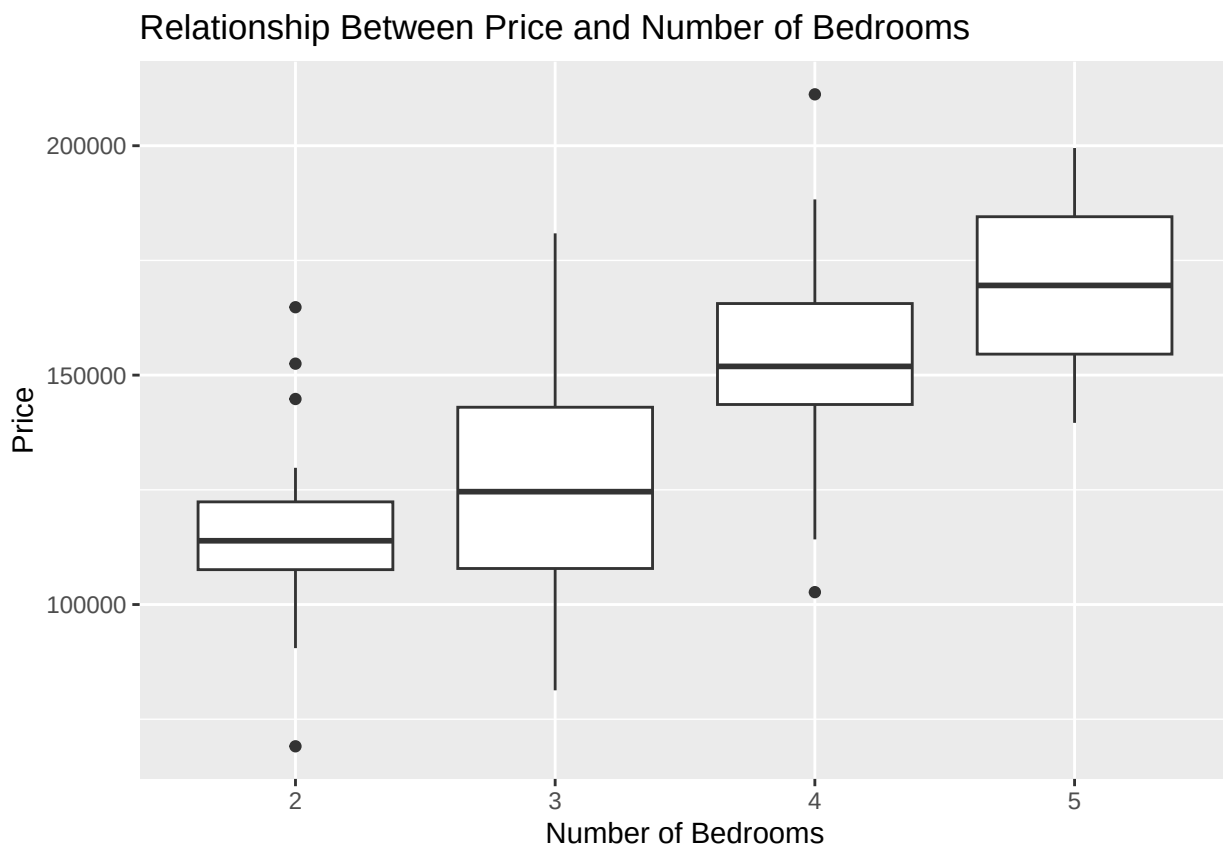

```
# I opted for a scatterplot as I believe this is the best plot to identify
# linear relationships. Observing the line of best fit compared to the points,
# there seems to be a weak linear relationship between price and size of the
# house
```

```
# Relationship Between price and number of bathrooms

bathPlot <- ggplot(house_df, aes(x = (as.factor(house_df$Bathrooms)), y = Price)) +
    geom_boxplot() + xlab("Number of Bathrooms") + ggtitle("Relationship Between Price and Number of Ba
bathPlot
```




```
# I opted for a boxplot to visualize this relationship due to little variance
# in x-values. This better organizes the data to visualize it. While box plots
# may not reveal linear relationship, it can be seen that price does increase
# with number of bathrooms.
```

```
# Relationship Between price and number of bedrooms

bedPlot <- ggplot(house_df, aes(x = (as.factor(house_df$Bedrooms)), y = Price)) +
    geom_boxplot() + xlab("Number of Bedrooms") + ggtitle("Relationship Between Price and Number of Bedi
bedPlot
```




```
# I opted for a boxplot to visualize this relationship due to little variance
# in x-values. This better organizes the data to visualize it. While box plots
# may not reveal linear relationship, it can be seen that price does increase
# with number of bedrooms. It does appear as if the medians of the data follow
# a linear relationship.
```

F. Next, create a simple regression model predicting price based on size, using the lm( ) command. In a comment, report the coefficient (aka slope or beta weight) of size in the regression output, and if it is statistically significant, interpret it with respect to price. Report the adjusted R-squared of the model and try to explain what it means.

```
lmSize <- lm(Price ~ SqFt, data = house_df)
summary(lmSize)
```

```
##
## Call:
## lm(formula = Price ~ SqFt, data = house_df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -46593 -16644  -1610  15124  54829
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -10091.130  18966.104  -0.532    0.596
```

```
## SqFt            70.226      9.426   7.450  1.3e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22480 on 126 degrees of freedom
## Multiple R-squared:  0.3058, Adjusted R-squared:  0.3003
## F-statistic:  55.5 on 1 and 126 DF,  p-value: 1.302e-11
```

```r
# Comment: The coefficient of size is 70.226. A P-value of 1.302e-11 shows
# statistical significance. This means size does influence price. The adjusted
# R-squared of 0.3003 implies a weak linear relationship.
```

G. Create a multiple regression model predicting price based on size AND the number of Bedrooms. Ensure that both predictors are incorporated into a single model, rather than creating separate models for each predictor. This will allow us to assess the relationship between each predictor and the outcome variable while controlling for the effect of the other predictor.

```r
lmSizeBed <- lm(Price ~ SqFt + Bedrooms, data = house_df)
summary(lmSizeBed)
```

```
##
## Call:
## lm(formula = Price ~ SqFt + Bedrooms, data = house_df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -45993 -15402  -1446  11041  48352
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6367.60   17827.91  -0.357    0.722
## SqFt           49.50      10.11   4.895 2.97e-06 ***
## Bedrooms    12486.06    2947.13   4.237 4.36e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21100 on 125 degrees of freedom
## Multiple R-squared:  0.393,  Adjusted R-squared:  0.3832
## F-statistic: 40.46 on 2 and 125 DF,  p-value: 2.826e-14
```

H. Report the **coefficients** of each predictor that is statistically significant and interpret them in a comment.

How does the *adjusted R-squared** compare to the adjusted R-squared from Step F? Is this better or worse?

```r
# Comment: Both size and number of bedrooms are marked as significant. The
# coefficient for size is 49.50, and for bedrooms it is 12486.06. Price
# increases more rapidly per added bedroom. The increase in price per square
# foot is more gradual. The adjusted R-squared of 0.3832 is greater than the
# one from Step F, but the relationship is still weak.
```

I. Use this code to create a one-row data frame with a single observation:

```
predict_df <- data.frame(SqFt = 1200, Bedrooms = 3)
```

Apply the **predict( )** function to this data frame to predict the **expected value of price**:

```
predict(lmSizeBed, predict_df)
```

```
##        1
## 90489.21
```

Can you assess the accuracy of the prediction using the model output from G? Explain in a comment.

```
# The model output of G provides an R-sqaured value of 0.393. This says 39.3%
# of the data can be predicted by this model. This means its likely the model
# might not output an accurate prediction
```

J. Create a new **multiple regression model**, with **Price** as the **outcome variable**, and **all four predictors** added to the model.

```
lmAll <- lm(Price ~ SqFt + Bedrooms + Bathrooms + Brick, data = house_df)
summary(lmAll)
```

```
##
## Call:
## lm(formula = Price ~ SqFt + Bedrooms + Bathrooms + Brick, data = house_df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -44438 -12879    618  14124  45344
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -27188.542  15264.772  -1.781 0.077359 .
## SqFt            35.801      9.241   3.874 0.000173 ***
## Bedrooms     10873.060   2523.693   4.308 3.33e-05 ***
## Bathrooms     9818.430   3699.285   2.654 0.009002 **
## Brick        21909.107   3371.179   6.499 1.81e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17630 on 123 degrees of freedom
## Multiple R-squared:  0.5828, Adjusted R-squared:  0.5692
## F-statistic: 42.96 on 4 and 123 DF,  p-value: < 2.2e-16
```

K. **Analysis**: Interpret the significant coefficients of the new model you created and review the quality of the model by commenting on its **adjusted R-Squared**. Based on the work you have done so far, including the visualizations and regression models you created, do you think there is a relationship between price and the predictor variables?

```
# Comment: The significant coefficients are size, number of bedrooms, and use
# of brick. With respective coefficients of 35.801, 10873.060, and 21909.107.
# The use of brick increases the price the most rapid, followed by number of
# bedrooms, and then square footage.

# An adgusted R-Squared value of 0.5692 suggests a moderately strong
# relationship.

# Based on the visualizations and regression models, I believe there is a
# relationship between price and the predictor variables.
```